Isolated (Spectral Garden)
Advanced Multi-Band Drum Isolation & Frequency Sweep Processor

Isolated is a specialized real-time audio plugin designed for surgical drum separation, transient management, and dynamic frequency sweeping. Powered by multi-stage transient splitters (TSS) and specialized channel frequency cross-overs, it allows producers to independently isolate, mix, and shape individual drum elements (Kick, Snare, Hats, Percussion, and Other) from a full stereo mix.

Core DSP Architecture & Signal Processing

1. **Multi-Band Crossover Engine (`channelFilters`)**: Splits incoming audio channels across dedicated frequency bands to segment percussion types.
2. **Transient/Sustain Splitter (TSS)**: Separates transient hits from body sustain per band, allowing independent manipulation of transient attack and release properties.
3. **Non-Linear Saturator & Envelope Tracker**: Adds character and analog warmth based on real-time dynamic envelope profiles.
4. **Synchronized Sidechain Ducking Engine**: Features both classic amplitude envelope ducking and DAW-tempo synchronized sidechain pump rates (1/4, 1/8, 1/16, 1/2 Beat).
5. **Dynamic Sweep Filter**: Includes a sweeping filter with adjustable Q and dynamic depth that reacts dynamically to transient energy peaks.

🎛️ Control Parameter Reference

Dynamic Channel Levels (Mix Matrix)

Control	Range	Default	Description
Kick Level (`ID_KICK`)	`0.0` to `1.0`	`0.8`	Adjusts the isolated gain contribution of kick frequencies.
Snare Level (`ID_SNARE`)	`0.0` to `1.0`	`0.8`	Adjusts the isolated gain contribution of snare elements.
Hats Level (`ID_HATS`)	`0.0` to `1.0`	`0.8`	Adjusts the isolated gain contribution of high-hats/cymbals.
Perc Level (`ID_PERC`)	`0.0` to `1.0`	`0.8`	Adjusts the isolated gain contribution of auxiliary percussive hits.
Other Level (`ID_OTHER`)	`0.0` to `1.0`	`0.5`	Gain scalar for residual or remaining non-drum spectral material.

Sweep & Filter Controls

Control	Range / Options	Default	Description
Sweep Frequency (`ID_SWEEP_FREQ`)	`20.0` to `20,000.0 Hz`	`1000.0 Hz`	Skew-mapped filter cutoff or sweep center frequency.
Sweep Q (`ID_SWEEP_Q`)	`0.1` to `8.0`	`1.0`	Resonance factor of the sweep filter.

| **Sweep Dynamic Depth** (`ID_SWEEP_DEPTH`) | `0.0` to `1.0` | `0.5` | Intensity of how transient energy modulates the filter sweep behavior. |
| **Isolate / Remove Mode** (`ID_SWEEP_MODE`) | Boolean (True/False) | True (Isolate) | Toggles between isolating the selected frequency region or notching/removing it. |

Dynamics & Transient Shaper Controls
Control	Range	Default	Description
Isolation Threshold (`ID_ISOLATION`)	`0.0` to `1.0`	`0.15`	Sensitivity threshold for gating and separating target signals.
Character Saturation (`ID_CHARACTER`)	`0.0` to `1.0`	`0.10`	Injects harmonic saturation based on real-time envelope tracking.
Transient Attack (`ID_TRANS_ATTACK`)	`0.5` to `15.0 ms`	`1.5 ms`	Attack time constant for the transient/sustain separator.
Transient Release (`ID_TRANS_RELEASE`)	`10.0` to `300.0 ms`	`45.0 ms`	Release time constant for the transient/sustain separator.

Sidechain & Global Utility Controls
Control	Range / Options	Default	Description
Sidechain Sync Rate (`ID_SC_SYNC_RATE`)	Off (Envelope), 1/4, 1/8, 1/16, 1/2 Beat	Off	Tempo-synced LFO pump rate ducking the rhythm section.
Sidechain Offset (`ID_SC_OFFSET`)	`-50.0` to `+50.0`	`0.0`	Time offset adjustment for sidechain ducking triggers.
Mix (`ID_MIX`)	`0.0` to `100.0%`	`100.0%`	Parallel dry/wet crossfade balance.
Bypass (`ID_BYPASS`)	Boolean (True/False)	False	Hard bypasses processor DSP path.

💾 State Management & Serialization
* **APVTS Framework**: Uses JUCE's robust `AudioProcessorValueTreeState` architecture to handle parameter automation and host sync.
* **State Recall**: Fully supports chunk-based preset saving and loading via standard XML serialization (`getStateInformation` / `setStateInformation`).

💻 Compilation Requirements
* **Framework**: JUCE 8.0+
* **Build Target**: VST3 (Windows x64)
* **C++ Standard**: C++20